Assignment 3

1. **(20 points)** Principal components analysis (PCA) reduces the dimensionality of the data by finding projection direction(s) that minimizes the squared errors in reconstructing the original data or equivalently maximizes the variance of the projected data. On the other hand, Fisher's linear discriminant is a supervised dimension reduction method, which, given labels of the data, finds the projection direction that maximizes the between-class variance relative to the within-class variance of the projected data.

   a. (10 points) In the following Figure 1, draw the first principal component direction in the left figure, and the first Fisher's linear discriminant direction in the right figure. Note: for PCA, ignore the fact that points are labeled (as round, diamond or square) since PCA does not use label information. For linear discriminant, consider round points as the positive class, and both diamond and square points as the negative class.

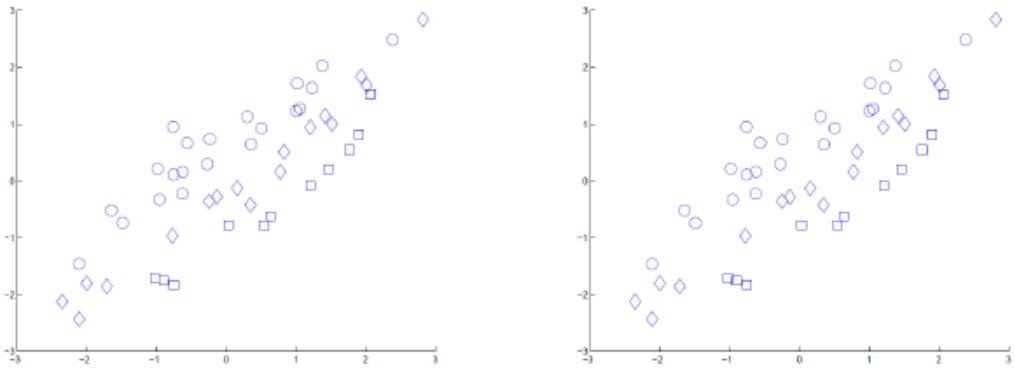


Figure 1: Draw the first principal component and linear discriminant component, respectively

b. (10 points) Consider 3 data points in the 2-d space: (-1, -1), (0,0), (1,1). What is the first principal component (write down the actual vector)? If we project the original data points into the 1-d subspace by the principal component you choose, what are their coordinates in the 1-d subspace? And what is the variance of the projected data? For the projected data you just obtained above, now if we represent them in the original 2-d space and consider them as the reconstruction of the original data points, what is the reconstruction error?

2. **(30 points)** K-means on MNIST. The MNIST dataset is a famous dataset in computer vision, which is available at `https://www.kaggle.com/datasets/oddrationale/mnist-in-csv`. The data files mnist_train.csv and mnist_test.csv contain gray-scale images of hand-drawn digits, from 0 through 9.

   Each image is 28 pixels in height and 28 pixels in width, for a total of 784 pixels in total. Each pixel has a single pixel-value associated with it, indicating the lightness or darkness of that pixel, with higher numbers meaning darker. This pixel-value is an integer between 0 and 255, inclusive.

   Let's consider the training data only for digits 0, 5, 8, 9.

   a. (10 points) Without using the label information, implement the K-means clustering for $K = 4$. It is up to you to get the best K-means algorithm for clustering this dataset. Draw these 4 centroids.

   b. (10 points) Analyzing the Performance in [a]. The images in the MNIST dataset have been labelled with the digit they actually represent. So, one way to evaluate our algorithm would be to examine the accuracy of the classification of the images from MNIST. For example, how many of the images known to be (labelled as) the digit 8 ended up in the centroid we think of as

representing the digit 8?

First, let's define the "classifications" of our data points provided by K-means. We take a majority-takes-all approach: we assign each cluster a label as the majority labels of all data points in this cluster. We define the accuracy of the classification for a centroid as:

$$\text{accuracy} = \frac{\#\text{ of majority labels}}{\#\text{ labels}}$$

We can repeat this step for all clusters and derive the accuracy measure of the algorithm. For every cluster, we first figure out what the majority label is for that cluster, and keep track of the count of the majority labels. The accuracy of the algorithm is then defined as:

$$\text{accuracy} = \frac{\text{sum of count of majority labels in all clusters}}{\text{total number of labels}}$$

Your score will depend on your final accuracy among all students' submissions.

c. (10 points) Provide a method to select K. For this problem, the true K is 4.

3. **(20 points)** Monte Carlo methods for Gaussian mixtures. A Gaussian mixture model is a family of distributions whose pdf is in the following form:

$$f(x) := \sum_{k=1}^{K} p_k \, \mathcal{N}(x|\mu_{\mathbf{k}}, \Sigma_k),$$

where $\mathcal{N}(\cdot|\mu, \Sigma)$ denotes the Gaussian pdf with mean μ and covariance Σ, and $\{p_1, \ldots, p_K\}$ are mixtuer weights satisfying

$$\sum_{k=1}^{K} p_k = 1, \quad p_k \geq 0, \quad k \in \{1, \ldots, K\}.$$

Consider a mixture of 8 Gaussians with means $\{(2 \cdot \cos\frac{2\pi k}{8}, 2 \cdot \sin\frac{2\pi k}{8}) | k = 0, \ldots 7\}$ and standard deviation 0.02. So $\mu_{\mathbf{k}} = (2 \cdot \cos\frac{2\pi k}{8}, 2 \cdot \sin\frac{2\pi k}{8}) \in \mathbb{R}^2$

and $\Sigma_k = 0.02^2 I_2 \in \mathbb{R}^{2\times 2}$ for $k = 0, \ldots, 7$. The random samples are shown in Figure 2.

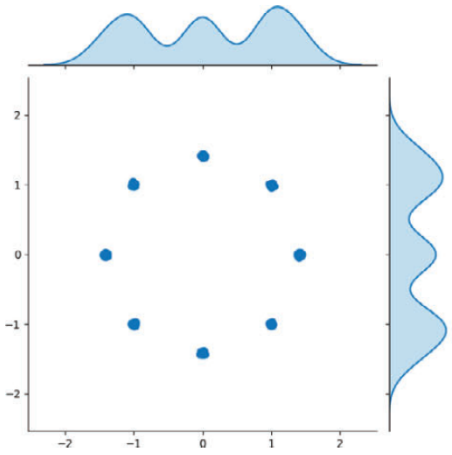


Figure 2: Gaussian mixtures

Without using the normality information and only using the formula of the pdf of Gaussian mixtures, implement one Monte Carlo method to obtain 800 samples from this distribution. Note that, if we use the normality information, this is a trivial question.

Since we know the true underlying distribution, we can provide a method to measure your performance in Part (a). Maximum mean discrepancy (MMD) is a kernel based statistical test used to determine whether given two distribution are the same. The code is availabe at `https://www.kaggle.com/code/onurtunali/maximum-mean-discrepancy/notebook`. Report the MMD between your simulated samples and real samples. The real samples can be trivially obtained by generating 100 samples from each bivariate normal distribution.

Your score will depend on your sample performance among all students' submissions.